(TR-104) PYTHON WITH AI/ML

Training Day 45 Report

Date: 13 March 2026

Topic Covered: RAG-Style Fine-Tuning, Context-Aware Training & Hallucination Reduction

The forty-fifth day of training focused on redesigning the fine-tuning approach for the RAG-based AI Assistant by creating context-aware datasets, improving instruction-following capability, and analyzing how fine-tuned models interact with retrieval pipelines.

- Identified the root cause of hallucination in the fine-tuned 3B model and observed that the model ignored retrieved RAG context during inference.
- Analyzed how training on plain question-answer pairs caused the model to memorize answers instead of using retrieved contextual information.
- Redesigned the training workflow by shifting from standard Q&A training to RAG-style context-aware fine-tuning.
- Built a new dataset structure where each training example included company context, user question, and expected answer.
- Created a RAG-style dataset containing multiple examples covering products, pricing, timings, support, refund policy, shipping, mission, security, and payment information.
- Added multiple wording variations for the same questions to improve robustness against unseen user phrasing patterns.
- Configured advanced LoRA training settings including higher LoRA rank, increased alpha value, and larger sequence length for handling long context inputs.
- Retrained the LLaMA 3.2 3B model on the context-aware dataset using LoRA adapters and optimized hyperparameters.
- Achieved significantly lower training loss after training with the redesigned dataset structure.
- Tested the retrained model on multiple company-related queries and verified improved response accuracy when context was directly provided.
- Exported the retrained model into GGUF format using quantization for efficient local deployment.

- Integrated the retrained model with the existing FastAPI-based RAG pipeline using Ollama.
- Performed end-to-end testing of retrieval, context injection, and answer generation workflow.
- Discovered that the original base LLaMA 3.2 model followed RAG instructions more effectively than the aggressively fine-tuned model during live retrieval-based inference.
- Strengthened understanding of context-aware fine-tuning, instruction following, hallucination behavior, retrieval grounding, and practical limitations of fine-tuning in RAG systems.
- GitHub Link: https://github.com/JaspinderKaurWalia26/finetuning/blob/main/fine_tuning.py
- Blog Link: https://medium.com/@jaspinderkaurwalia855/building-a-reliable-ai-assistant-with-rag-and-fine-tuned-llama-3-2-dc01a0919835